

PFE FX Simulation Model Validation

Deterministic FX Drift and Path-Wise Arbitrage

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Current Model Specification: FX Drift

The Theoretical Baseline:

The deterministic component $g_k^{FX}(t)$ acts as the mean function, since $E[Y_k^{FX}(t)] = g_k^{FX}(t)$.

This drift is specified strictly by the FX forward curve implied by the interest rate curves of the two currencies at the initial date of simulation ($t = 0$).

Assuming the rate is quoted as USD per unit of foreign currency, the drift is calculated via interest rate parity:

$$g_k^{FX}(t) = Y_k^{FX}(0) \frac{DF_{FOR}(0, t)}{DF_{USD}(0, t)}$$

where $DF_{FOR}(0, t)$ and $DF_{USD}(0, t)$ are the discount factors for the foreign currency and USD respectively.

The Core Limitation: Missing Cross-Gamma

The Disconnect:

- The model simulates Interest Rates stochastically using a shifted exponential Vasicek process, meaning rates evolve dynamically over time.
- However, the FX drift remains *locked* to the static $t = 0$ forward curve.

Why this fails: In an arbitrage-free simulation, the Covered Interest Parity (CIP) must hold at *every time step* on *every individual path*. By keeping the FX drift deterministic while rates are stochastic, the model ignores the compounding correlation between fluctuating interest rates and the FX spot rate (cross-gamma risk).

Path-Wise Failure Illustration: Setup at $t = 0$

Let us illustrate this failure on a specific simulated path (ω_1) for EUR/USD.

Market Conditions at $t = 0$:

- **Spot FX:** $Y^{FX}(0) = 1.00$ USD per EUR.
- **Interest Rates:** Both US and EUR have flat yield curves at 2%.
- **Model FX Drift:** Because rates are equal, the ratio of discount factors is 1. The model locks in an expected drift of $g^{FX}(t) = 1.00$ for all future times t .

Path-Wise Failure Illustration: Time $t = 1$

Fast forward to time $t = 1$ (year for instance) on path ω_1 .

Stochastic Rates Diverge:

- The IR simulation generates a severe US rate cut: The 1Y US zero rate is now **0%**.
- The IR simulation generates a EUR rate hike: The 1Y EUR zero rate is now **10%**.

Static FX Spot:

- Because the FX drift was deterministically locked at $t = 0$, the FX spot $Y^{FX}(1)$ is completely unaware of this rate divergence.
- Assuming the random Wiener increment is zero for simplicity, the simulated spot sits exactly at its mean: **1.00 USD per EUR**.

Path-Wise Failure Illustration: The Arbitrage

At $t = 1$, the pricing engine inside the PFE model observes: Spot = 1.00, US Rate = 0%, EUR Rate = 10%.

Pricing a 1-Year Forward Contract at $t = 1$: To prevent static arbitrage at $t = 1$, the pricing engine calculates the 1-year forward rate (for maturity at $t = 2$):

$$F(1, 2) = 1.00 \times \frac{0.909}{1.000} = 0.909 \text{ USD per EUR}$$

The Simulation Paradox at $t = 2$:

- The forward contract was struck at **0.909**.
- But the simulation engine continues to drive the actual spot rate based on the $t = 0$ deterministic drift.
- At $t = 2$, the simulated spot rate arrives at **1.00**.

The Result: Fictitious Profits

- The forward contract consistently pays out $1.00 - 0.909 = 0.091$ USD at $t = 2$ on this path.
- The simulated portfolio shows a risk-free profit simply by holding the forward contract to maturity.

Impact on Exposure Metrics: This violates the fundamental theorem of asset pricing within the simulation framework. Cross-currency derivatives will be systematically mispriced along the paths, leading to highly inaccurate Mark-to-Market (MtM) distributions and fundamentally flawed Potential Future Exposure (PFE) metrics at the tail percentiles.

The BSHWWH (XVA Model) Benchmark : Coupled SDEs (RAG to confirm)

Dynamic vs. Deterministic Drift

The BSHWWH model resolves this by defining a coupled system of Stochastic Differential Equations (SDEs).

Instead of a locked $t = 0$ forward curve, the FX drift is forced to equal the instantaneous difference between the simulated domestic and foreign short rates under the risk-neutral measure:

$$\mu_{FX}(t) = r_d(t) - r_f(t)$$

The Coupled FX Spot Process:

$$dS(t) = (r_d(t) - r_f(t)) S(t)dt + \sigma_S S(t)dW_S(t)$$

This ensures that the exchange rate $S(t)$ is mathematically bound to the stochastic interest rate environment on *every single simulated path*.

Revisiting the Path under BSHWW: Time $t = 1$

Let's revisit our path ω_1 where rates diverged at $t = 1$.

Stochastic Rates Diverge (Same as Before):

- Simulated 1Y US Rate (r_d): **0%**
- Simulated 1Y EUR Rate (r_f): **10%**
- Spot rate at $t = 1$: **1.00**

Dynamic FX Drift Updates instantly:

- Because the SDEs are coupled, the FX simulation engine immediately updates the drift for the next time step:

$$\mu_{FX}(1) = 0\% - 10\% = -10\%$$

- The simulation engine now expects the EUR to depreciate by 10% over the next year to satisfy Covered Interest Parity.

Revisiting the Path under BSHWW: Time $t = 2$

At $t = 1$, the pricing engine prices the 1-year forward contract exactly as before.

$$F(1, 2) = 1.00 \times \frac{0.909}{1.000} = \mathbf{0.909}$$

The Simulation Engine Aligns:

- The BSHWW simulation engine drives the spot rate forward using the updated -10% drift.
- Assuming zero random innovation again, the simulated spot rate $S(2)$ arrives at exactly **0.909**.

Arbitrage Eliminated: The forward contract payout at $t = 2$ is now $0.909 - 0.909 = \mathbf{0.00}$. The simulated spot path and the pricing forward curve are in perfect mathematical harmony, preserving path-wise arbitrage-freeness for PFE calculations.

Monitoring: Path-Wise CIP Violation Tracking

The "Arbitrage Error" Metric (ϵ):

Build a sanity check into the engine to track the magnitude of the arbitrage on a sample of paths.

For a given future time t and maturity T , calculate the absolute difference between the implied forward rate and the simulation's expected spot rate:

$$\epsilon(t) = \left| S_{sim}(t) \frac{DF_{FOR,sim}(t, T)}{DF_{USD,sim}(t, T)} - g_k^{FX}(T) \right|$$

Actionable Threshold: If $\epsilon(t)$ exceeds a predefined tolerance (e.g., 50 bps) on a significant percentage of paths, flag the PFE distribution for that currency pair as compromised.